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Advanced ultrasound-assisted degradation of pharmaceutical wastewater: a sustainable solution for eutrophication control

N Hardyanti^{1*}, B Zaman¹, N F Putri¹, A B D Phitaloka¹, M B Wahyuarta¹, P Purwono²

¹Department of Environmental Engineering, Faculty of Engineering, Universitas Diponegoro, Semarang, Indonesia 50275

²Department of Environmental Science, UIN Raden Mas Said Surakarta, Jl. Pandawa, Pucangan, Kartasura, Indonesia 57168

E-mail: *nurandanihardyanti@gmail.com

Abstract. Waste from pharmaceutical production processes has the potential to pollute aquatic environments, especially the occurrence of eutrophication events. Environmental indicators that usually affect eutrophication include Phosphate, Sulfate, and Ammonia, which exceed the limits set by regulations. This study focuses on investigating the fluctuations in the quantity of wastewater contaminant parameters from pharmaceutical production facilities before and after treatment using ultrasonic methods with variations in time and pH and testing the effectiveness of the ultrasonic process on water quality using predetermined parameters. This study involved independent variables, including ultrasound time and pH variables. Time variations are divided into 30, 60, 90, 120 minutes. Meanwhile, pH variations are divided into pH 3, 5, 7, 9, and 12. The most effective results were in the 30-minute time variation and pH 9; this is indicated by the phosphate removal efficiency reaching 50.2% and 46.8 for each variation. Phosphate, in this case, is the main parameter because the primary nutrient for eutrophication is phosphate. Control variables include the amount of initial contaminants and instrument conditions.

1. Introduction

Pharmaceutical and Personal Care Products (PPCPs) encompass various categories of chemical substances, including pharmaceuticals and personal care items. PPCPs used by humans are typically discharged into the environment as waste after use, making their presence in aquatic systems a primary contemporary concern [1]. The PPCPs contain chemical compounds that are designed to be biologically active and react through specific mechanisms in the human body. As a result, concerns have arisen about the potential effects of these active PPCPs on the environment and both human health and ecosystems [2].

Eutrophication is a process of nutrient enrichment in water bodies, leading to excessive algal growth (algal blooms) and detrimental ecosystem changes [3], [4], [5]. Ammonia, sulfate, and phosphate are key nutrients that contribute to eutrophication when present in excessive amounts [56]. Ammonia (NH_3), a nitrogen compound produced from organic matter decomposition, promotes eutrophication as nitrogen is essential for algal and aquatic plant growth [8], [9], [10]. Sulfate (SO_4^{2-}) acts as an alternative to oxygen under anaerobic conditions, producing sulfide that binds with iron and releases phosphate from sediments [6,11–12]. Phosphate (PO_4^{3-}) is a critical limiting nutrient for algal growth in freshwater, so elevated phosphate levels directly stimulate algal blooms and eutrophication [3], [9].

Pharmaceutical wastewater is a significant contributor to eutrophication due to its complex mixture of pollutants, which are often difficult to treat. Ultrasonic technology emerging a promising



approach to addressing eutrophication by targeting its causative pollutants [13]. In recent years, numerous studies have demonstrated that Advanced Oxidation Processes (AOPs) are effective in degrading persistent and complex pollutant compounds. The ultrasound process operates on the principle of ultrasonic irradiation, which induces the sonolysis of water into $\bullet\text{H}$ and $\bullet\text{OH}$ radicals. Previous research has highlighted that increasing concentrations of nutrients, especially nitrogen and phosphorus concentration in water bodies lead to eutrophication, decreasing dissolved oxygen and harming the aquatic ecosystem. Therefore, integrating ultrasound treatment with biotechniques can remove nitrogen and phosphorus, resulting in a promising approach to eutrophication mitigation [14]. Ultrasound treatment (US) can control bacteria and algae in industrial water systems through shear forces, microbubbles, and high-frequency, low-power energy [15]. The physical stress induced by ultrasonic waves can disrupt sessile and planktonic biological populations, potentially triggering programmed cell death [15]. In summary, ultrasonic technology presents a multifaceted approach to eutrophication control, with the potential to disrupt algae, enhance treatment processes, and improve water quality [15].

2. Materials and methods

2.1 Pharmaceutical wastewater sample

The pharmaceutical wastewater sample was collected from a pharmaceutical industry in Semarang, Indonesia. The sampling occurred at the inlet point of the industry's wastewater treatment plant. The untreated wastewater contains ammonia at 0.096 mg/L, phosphate at 0.658 mg/L, and sulfate at 59.21 mg/L. The relatively high sulfate concentration, along with the presence of phosphate and ammonia, has the potential to degrade water quality, trigger eutrophication, and cause odor and corrosion if discharged without treatment.

2.2 Ultrasound process

Ultrasound technology utilizes cavitation bubbles, where these bubbles' formation, growth, and collapse generate intense localized energy [16–17]. This process creates extreme conditions, such as high temperatures and pressures, leading to the degradation of organic compounds [16]. The collapse of cavitation bubbles produces localized hot spots with temperatures reaching thousands of Kelvins, which can directly break down organic pollutants [16]. Ultrasound generates free radicals (e.g., $\bullet\text{OH}$) through water molecule dissociation, which then react with and degrade organic pollutants [18]. These free radicals oxidize organic contaminants, breaking them into simpler and less harmful substances [19].

2.3 Analysis method

This study conducted the Environmental parameter analysis following the ultrasound treatment. The analytical procedures were carried out in accordance with the applicable Indonesian National Standards (SNI). Ammonia was measured using the phenate spectrophotometric method based on SNI 06-6989.30-2005. In this procedure, ammonia reacts with hypochlorite and phenol, catalyzed by sodium nitroprusside, to form a blue indophenol complex. The intensity of the resulting color is measured spectrophotometrically at a wavelength of 640 nm. Sulfate was analyzed using the turbidimetric method under SNI 6989.20:2019. The principle of this analysis involves reacting sulfate ions with barium chloride in an acidic condition. This reaction forms a barium sulfate suspension, in which uniformly sized barium sulfate crystals are produced. The resulting turbidity is measured spectrophotometrically at a wavelength of 420 nm. Meantime, phosphate was determined by the ascorbic acid spectrophotometric method as outlined in SNI 06-6989.30-2005. Orthophosphate in an acidic medium reacts with ammonium molybdate and potassium antimonyl tartrate to form phosphomolybdic acid, which is subsequently reduced by ascorbic acid to produce a molybdenum blue complex. The absorbance of this complex is then measured spectrophotometrically at a wavelength of 880 nm. The removal percentage for each parameter was calculated using the following formula.

$$\text{Removal percentage} = \frac{c_0 - c_1}{c_0} \times 100\% \quad (1)$$

With c_0 as the initial concentration before the ultrasound treatment and c_1 as the final concentration after the ultrasound treatment.

3. Results and discussion

3.1 Effect on ammonia

The presence of ammonia in pharmaceutical wastewater poses considerable environmental and health concerns due to its toxicity and its role in promoting water eutrophication [20], [21], [22]. Ammonia is a prime component of the nitrogen cycle and a significant contributor to eutrophication [23]. Elevated ammonia concentrations can directly stimulate algal growth, exacerbating eutrophication [24–25]. Additionally, some forms of nitrogen, such as ammonia, can harm aquatic organisms in high concentrations. [26]. This study investigating four ultrasound exposure times, those were 30 minutes, 60 minutes, 90 minutes, and 120 minutes, to identify the optimal exposure time. Once the optimal duration was determined, five different pH level were applied: pH 3, pH 5, pH 7, pH 9, and pH 12.

Table 1. Ammonia test results

Variation	Ammonia	Treatment Efficiency
Initial	0.906 mg/L	
30 minutes	0.834 mg/L	-
60 minutes	0.957 mg/L	-
90 minutes	1.483 mg/L	-
120 minutes	2.022 mg/L	-
pH 3	1.602 mg/L	-
pH 5	1.132 mg/L	-
pH 7	1.123 mg/L	-
pH 9	0.745 mg/L	-
pH 12	1.834 mg/L	-

The expected outcome of the treatment was a reduction in ammonia concentration following ultrasonic irradiation with variations in exposure time and pH, starting from an initial ammonia level of 0.096 mg/L. However, the results showed an unexpected increase in ammonia levels, with the lowest increase observed at a 30-minute exposure time (0.834 mg/L) and pH 12 (0.745 mg/L). The highest increase was recorded at 120 minutes (2.002 mg/L) and pH 12 (1.834 mg/L).

This increase in ammonia concentration contrasts with findings from previous studies, which reported that ultrasonic treatment could enhance the removal of ammonia and other pollutants, either as a standalone method or in combination with other techniques [27–28]. Ultrasonic irradiation generates cavitation bubbles in the liquid medium, which rapidly expand and collapse, producing localized hotspots with extremely high temperatures and pressures [29]. Under such conditions, ammonia molecules inside the cavitation bubbles may undergo pyrolysis, breaking down into nitrogen and hydrogen gases [29]. However, the efficiency of this process is influenced by several factors, including the ultrasound frequency and power, the initial ammonia concentration, and the solution's pH [30–31]. Under certain conditions, ultrasonic treatment may instead lead to an increase in ammonia concentration [32]. Ultrasonic irradiation can break down complex organic compounds [32]. The sonolysis process produces hydroxyl radicals ($\bullet\text{OH}$), which react with organic pollutants [32]. Nevertheless, the breakdown of nitrogen-containing organic compounds leading to the release of ammonia into the solution, increasing ammonia concentration [32]. Furthermore, pharmaceutical wastewater is known for its complex composition, containing many organic substances [33]. Several studies have reported that during ultrasound treatment, the COD concentration may increase due to the

fragmentation of complex organic compounds into simpler forms [32]. In summary, ultrasound treatment can potentially elevate ammonia levels when decomposing nitrogenous organic compounds.

3.2 Effect on sulphate

Sulfate in environmental samples is of wide concern due to its potential impacts on air quality, balance, and human health [34]. Sulfate is an essential component in pharmaceutical wastewater, which arises from various production processes and raw materials [35]. Sulfate plays a significant, often underestimated, role in eutrophication. Elevated sulfate concentrations can enhance phosphorus mobilization, promote toxic conditions, and disrupt nutrient cycling in aquatic ecosystems [36–37]. In addition, elevated sulfate concentrations contribute to eutrophication by driving the release of phosphorus from sediments. This study used four variations of Ultrasound time, namely 30 minutes, 60 minutes, 90 minutes, and 120 minutes. After obtaining the optimal time variation, five pH variations were carried out, namely pH 3, pH 5, pH 7, pH 9, and pH 12. The results are obtained in the table below.

Table 2. Sulphate test results

Variation	Sulphate	Treatment Efficiency
Initial	59.21 mg/L	
30 minutes	98.811 mg/L	-
60 minutes	79.69 mg/L	-
90 minutes	76.311 mg/L	-
120 minutes	74.598 mg/L	-
pH 3	162.81 mg/L	-
pH 5	52.48 mg/L	11.4%
pH 7	102.47 mg/L	-
pH 9	57.87 mg/L	2.2%
pH 12	11.59 mg/L	80.42%

From the results in the table, the most optimal Ultrasound time variation among other variations was found at 120 minutes, but there was still an increase from the initial concentration of 59.21 mg/L to 74.598 mg/L. At the same time, the highest increase in the time variation was 30 minutes, with an increase from the original 59.21 mg/L to 98.811 mg/L. Thus, the optimum variation of Ultrasound time was obtained, and pH variation testing was carried out, pH 12 produced the best removal efficiency with a removal efficiency allowance of 80.42%, while the least optimum pH variation was obtained at pH 3 because there was an increase in sulfate concentration from 59.21 mg/L to 162.81 mg/L. This has been discussed and is in line with previous journals explaining that Ultrasound can increase the activity of sulfate-reducing bacteria (SRB) and increase the efficiency of sulfate removal [38]. Sulfate reduction by sulfate-reducing bacteria (SRB) is very sensitive to pH [39]. Very low pH inhibits the activity of sulfate-reducing bacteria (SRB); therefore, maintaining the appropriate pH is very important for sulfate reduction efficiency [39]. However, other studies have shown that the optimal pH for sulfate reduction by SRB is usually in the neutral to slightly acidic range. However, certain conditions may vary depending on the system used [40]. This differs from the existing test results, which show that high sulfate reduction results are influenced by high pH. Another study stated that under high pH conditions, phosphate removal can be enhanced using calcite [41]. Although this study did not directly address sulfate in pharmaceutical wastewater, it showed how high pH can affect the removal of other ions, impacting sulfate behavior. Additional technologies, such as sulfate radical-based high-level oxidation (SR-AOP) technology, VUV/UV/Fenton, can effectively treat pharmaceutical wastewater at near-neutral pH [42]. With the results and several theories above, ultrasound testing with pH can affect sulfate results; acidic pH causes sulfate degrading bacteria (SRB) to not be in optimal conditions, so variations in neutral and alkaline are good enough to reduce sulfate content in pharmaceutical waste.

3.3 Effect on phosphate

Phosphorus is essential for all life forms and often acts as a limiting nutrient in aquatic ecosystems [43–44]. Increased phosphate levels can stimulate algal and cyanobacterial growth [45]. As these blooms die and decompose, bacterial activity intensifies, depleting dissolved oxygen and creating anoxic conditions that endanger aquatic life [45]. Phosphate is a key parameter that can contaminate aquatic environments, as it contributes to eutrophication and oxygen depletion in aquatic environments [46]. The phosphate content in wastewater consists of both organic phosphorus (OP) and inorganic phosphate (IP) [47]. This study investigating four ultrasound exposure times, those were 30 minutes, 60 minutes, 90 minutes, and 120 minutes, to identify the optimal exposure time. Once the optimal duration was determined, five different pH level were applied: pH 3, pH 5, pH 7, pH 9, and pH 12.

Table 3. Phosphate test results

Variation	Phosphate	Treatment Efficiency
Initial	0.658 mg/L	
30 minutes	0.315 mg/L	50.1%
60 minutes	0.398 mg/L	38.1%
90 minutes	0.333 mg/L	47.1%
120 minutes	0.369 mg/L	40.8%
pH 3	0.969 mg/L	-
pH 5	0.679 mg/L	-
pH 7	0.443 mg/L	32.7%
pH 9	0.350 mg/L	46.8%
pH 12	0.964 mg/L	-

After treating a water sample that started with 0.658 mg/L of phosphate, we expected the phosphate levels to decrease when we changed the time and pH during the ultrasound treatment. The most effective phosphate removal occurred with a sonication time of 30 minutes, resulting in a concentration of 0.315 mg/L and a removal percentage of 50.1%. The lowest removal was observed at 60 minutes, with a concentration of 0.398 mg/L and a removal efficiency of 38.1%. For the pH variation treatment, the most optimal phosphate removal occurred at pH 9, resulting in a concentration of 0.350 mg/L and a removal efficiency of 46.8%. In contrast, the highest increase in phosphate concentration was observed at pH 3, where the level rose from 0.658 mg/L to 0.969 mg/L. This aligns with previous research indicating that ultrasound can generate cavitation bubbles, which produce extreme temperatures and pressures capable of breaking down complex organic molecules such as phosphate [48–49]. Ultrasound enhances the degradation of organic matter containing dissolved phosphate. The breakdown of these organic compounds can release phosphate ions, making them easier to remove in subsequent treatment processes [50–51]. These statements suggest that ultrasound exposure time and pH are critical factors influencing the effectiveness of phosphate removal in ultrasound-assisted treatment processes.

4. Conclusion

This study aims to reduce pharmaceutical wastewater pollutants by testing the effectiveness of ultrasound treatment with variations in time and pH. Ultrasound technology utilizes mechanical, thermal, and chemical effects derived from its cavitation effect. This method approach is modeled into variations in time and pH. A series of tests showed that the best variation of phosphate levels was shown at a time variation of 30 minutes and pH 9; this was indicated by the phosphate removal efficiency reaching 50.2% and 46.8 for each variation. Phosphate, in this case, is the main parameter because the primary nutrient for eutrophication is phosphate. At the same time, the best variation for the sulfate parameter was at a pH variation of 12, with a removal efficiency of 80.42% from the initial parameter. Unfortunately, there was an undesirable increase for all variations in the ammonia parameter. From this study, the ultrasound method is quite influential in improving water quality. Still,

it will be more optimal if integrated with other methods such as Fenton, electrocatalysis, etc. Further studies to improve the ultrasound process can potentially produce significant breakthroughs for environmental and industrial applications.

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